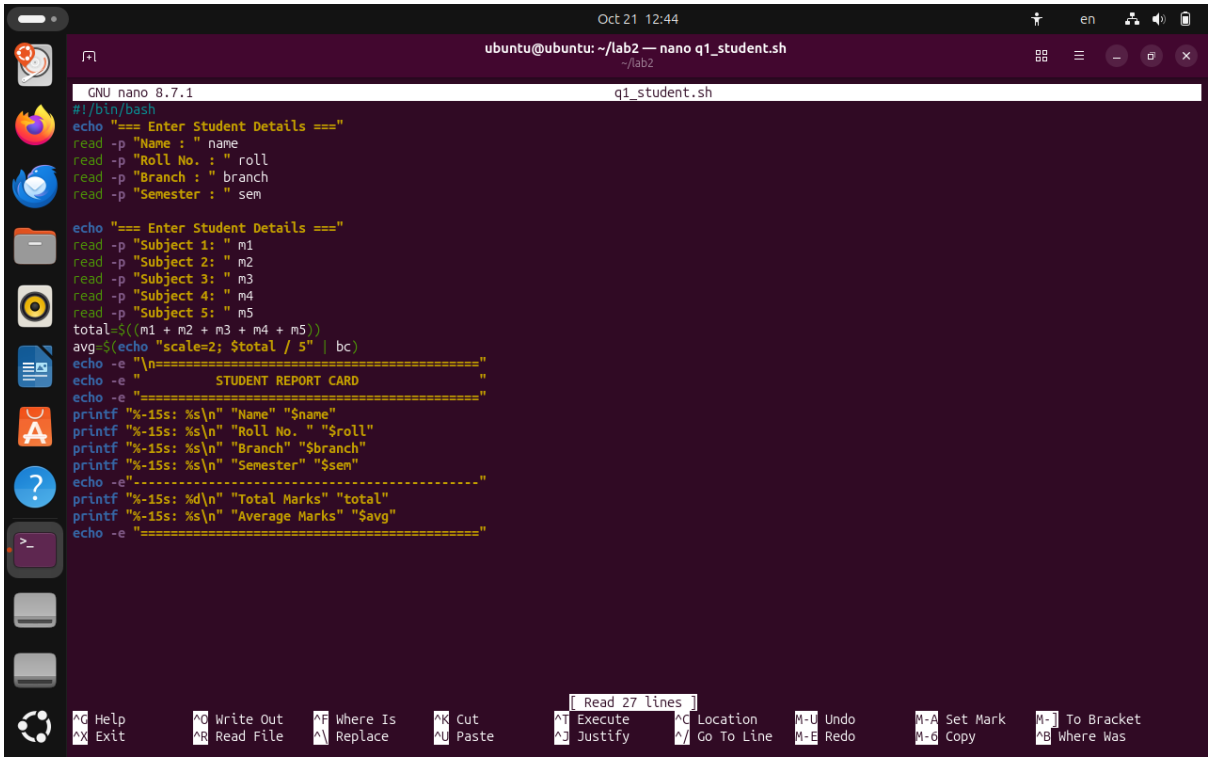
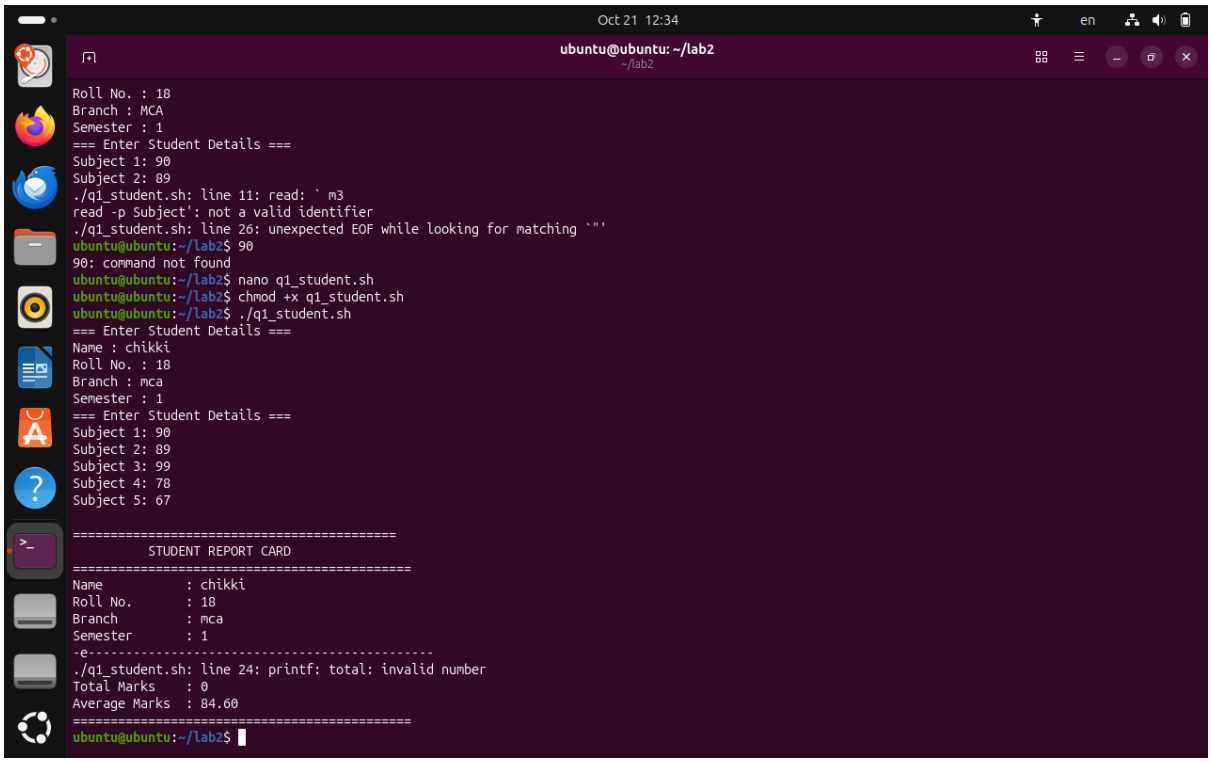
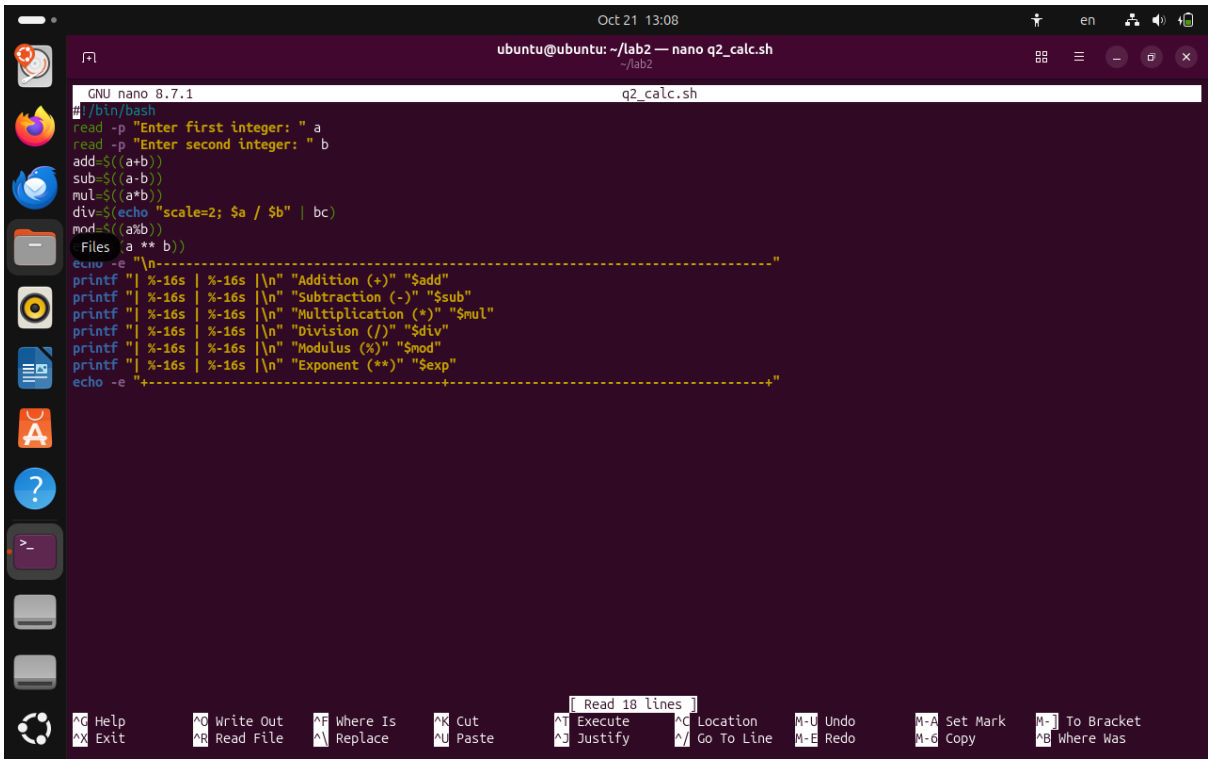
Ques1-Student Information System script accepting Name, Roll No., Branch, Semester and marks in five subjects using variables; display a formatted report.



Output:-



Ques2-. Implement arithmetic operations in shell scripts.



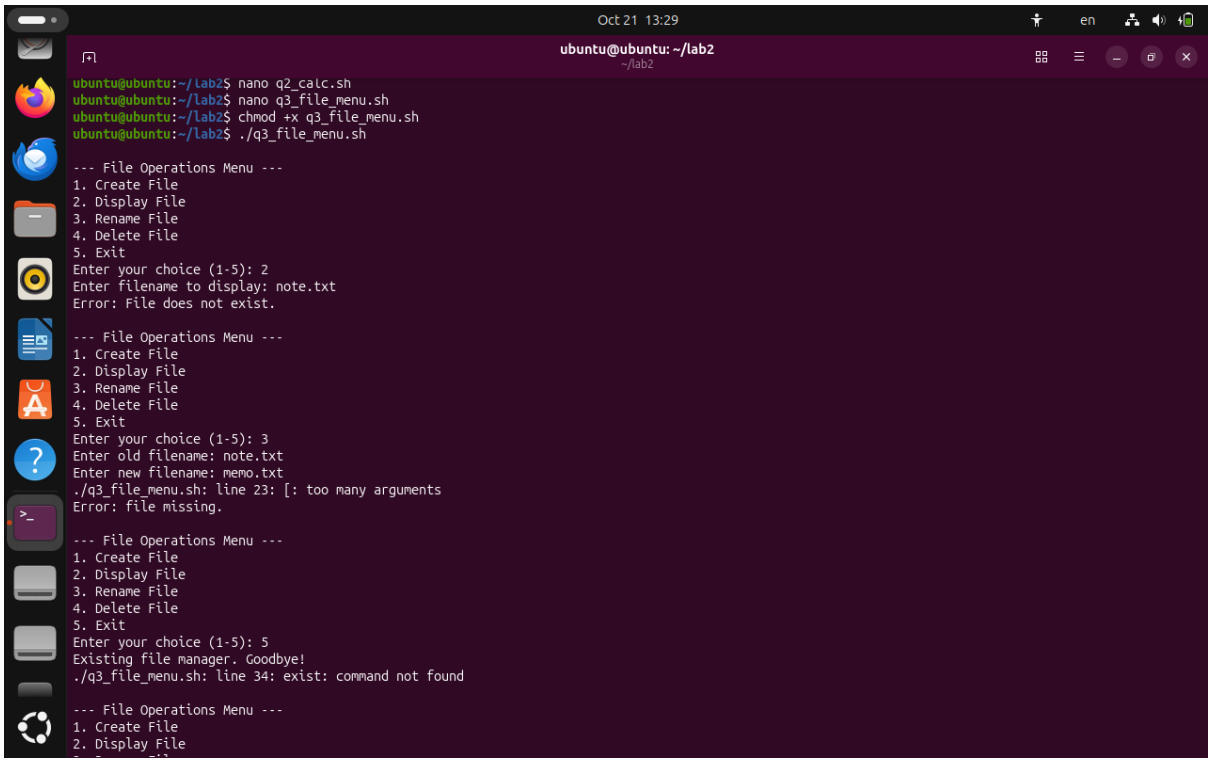
Output:-



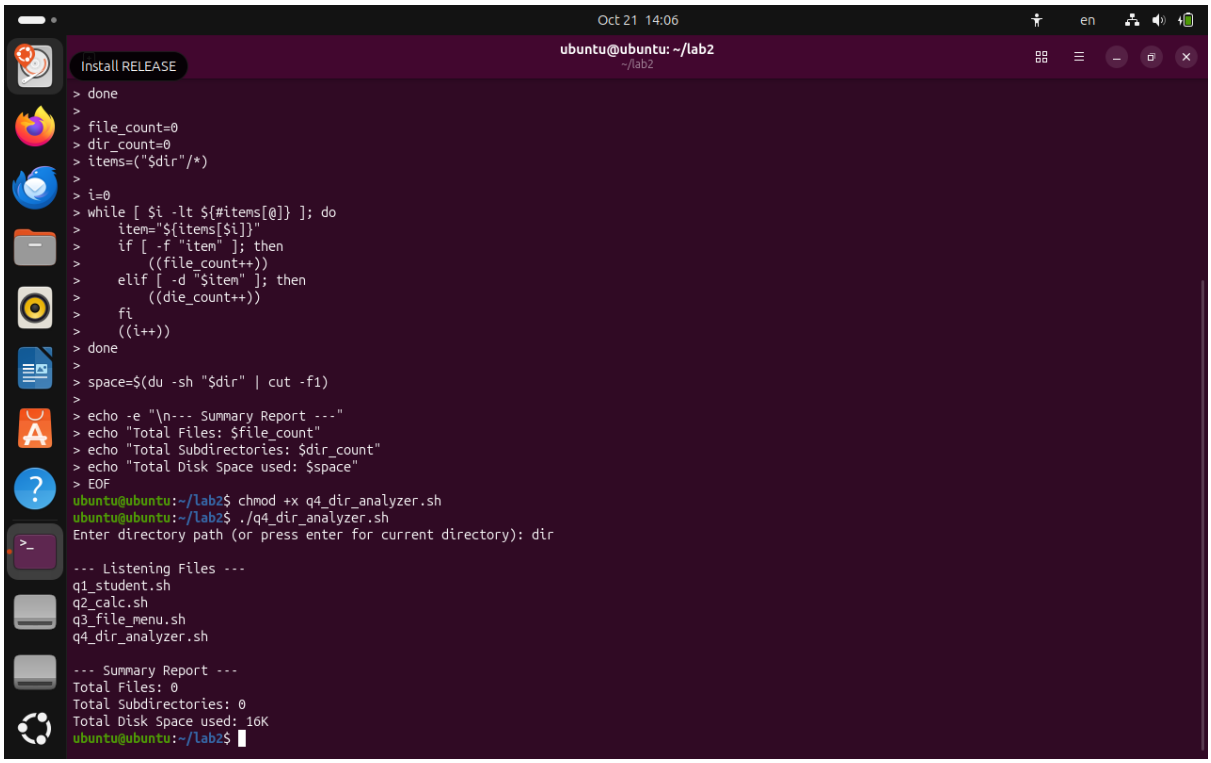
Ques3-Menu-Driven File Operations Using case



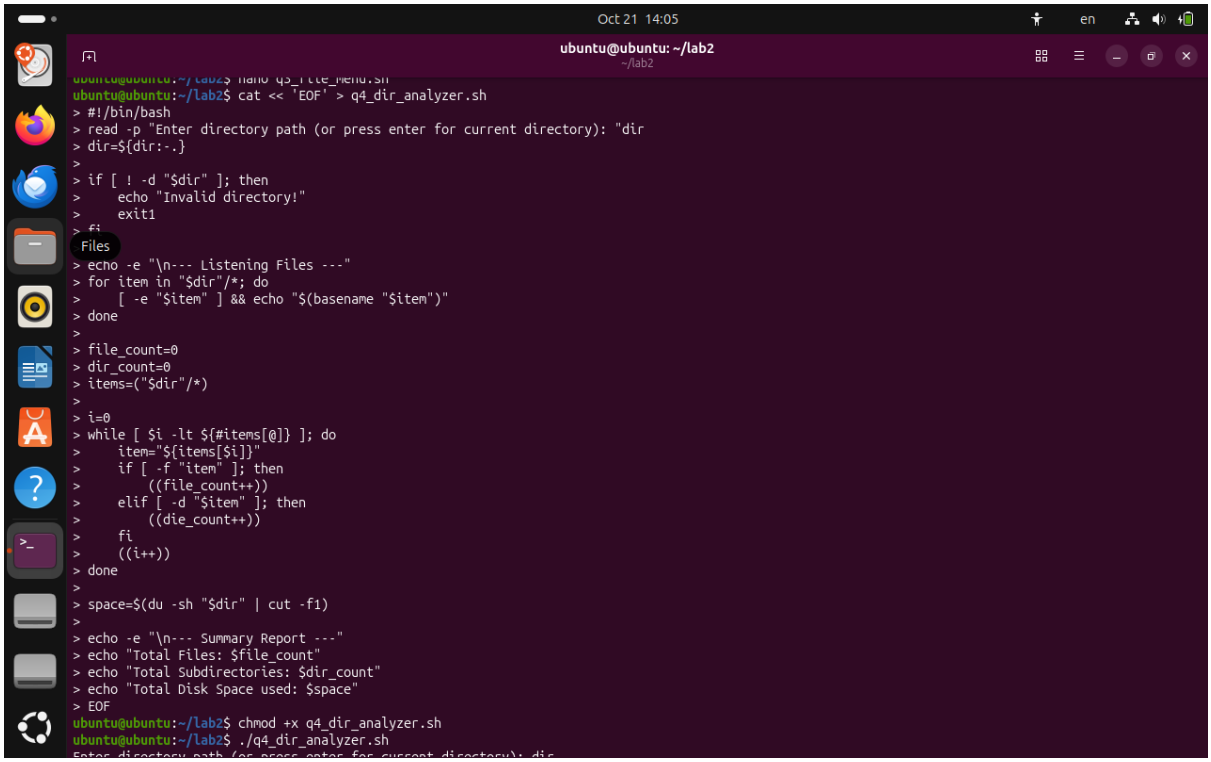
Output:-



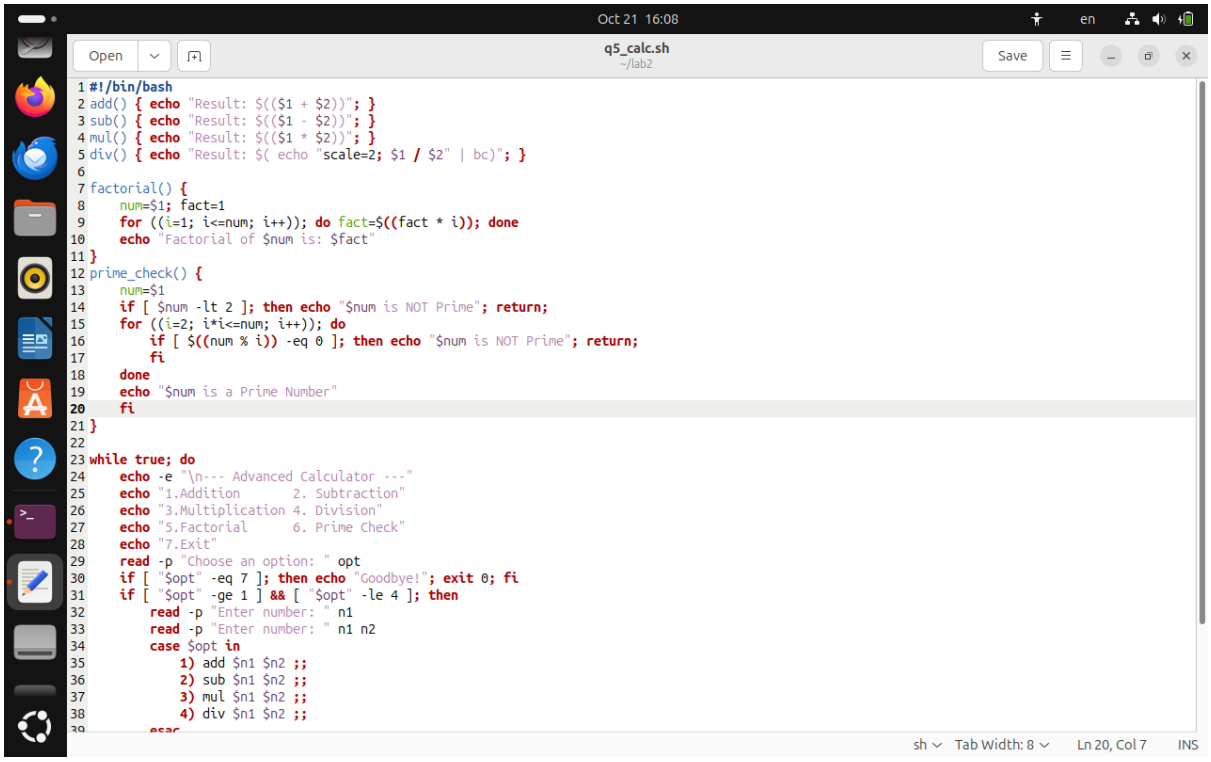
Ques4- Directory Listing and Disk Usage Using Loops.


Oct 21 14:06
en
ubuntu@ubuntu: ~/lab2
~/lab2
Install RELEASE
> done
>
> file_count=0
> dir_count=0
> items=("$dir"/*)
>
> i=0
> while [$i -lt ${#items[@]}]; do
>   item="${items[$i]}"
>   if [-f "item"]; then
>   ((file_count++))
>   elif [-d "$item"]; then
>   ((die_count++))
>   fi
>   ((i++))
> done
>
> space=$(du -sh "$dir" | cut -f1)
>
> echo -e "\n--- Summary Report ---"
> echo "Total Files: $file_count"
> echo "Total Subdirectories: $dir_count"
> echo "Total Disk Space used: $space"
> EOF
ubuntu@ubuntu:~/lab2$ chmod +x q4_dir_analyzer.sh
ubuntu@ubuntu:~/lab2$ ./q4_dir_analyzer.sh
Enter directory path (or press enter for current directory): dir
--- Listening Files ---
q1_student.sh
q2_calc.sh
q3_file_menu.sh
q4_dir_analyzer.sh
--- Summary Report ---
Total Files: 0
Total Subdirectories: 0
Total Disk Space used: 16K
ubuntu@ubuntu:~/lab2$

Output:-


Oct 21 14:05
en
ubuntu@ubuntu: ~/lab2
~/lab2
ubuntu@ubuntu:~/lab2$ cat << 'EOF' > q4_dir_analyzer.sh
> #!/bin/bash
> read -p "Enter directory path (or press enter for current directory): "dir
> dir=${dir:-.}
>
> if [! -d "$dir"]; then
>   echo "Invalid directory!"
>   exit1
> fi
Files
> echo -e "\n--- Listening Files ---"
> for item in "$dir"/*; do
>   [-e "$item"] && echo "$(basename "$item")"
> done
>
> file_count=0
> dir_count=0
> items=("$dir"/*)
>
> i=0
> while [$i -lt ${#items[@]}]; do
>   item="${items[$i]}"
>   if [-f "item"]; then
>   ((file_count++))
>   elif [-d "$item"]; then
>   ((die_count++))
>   fi
>   ((i++))
> done
>
> space=$(du -sh "$dir" | cut -f1)
>
> echo -e "\n--- Summary Report ---"
> echo "Total Files: $file_count"
> echo "Total Subdirectories: $dir_count"
> echo "Total Disk Space used: $space"
> EOF
ubuntu@ubuntu:~/lab2$ chmod +x q4_dir_analyzer.sh
ubuntu@ubuntu:~/lab2$ ./q4_dir_analyzer.sh

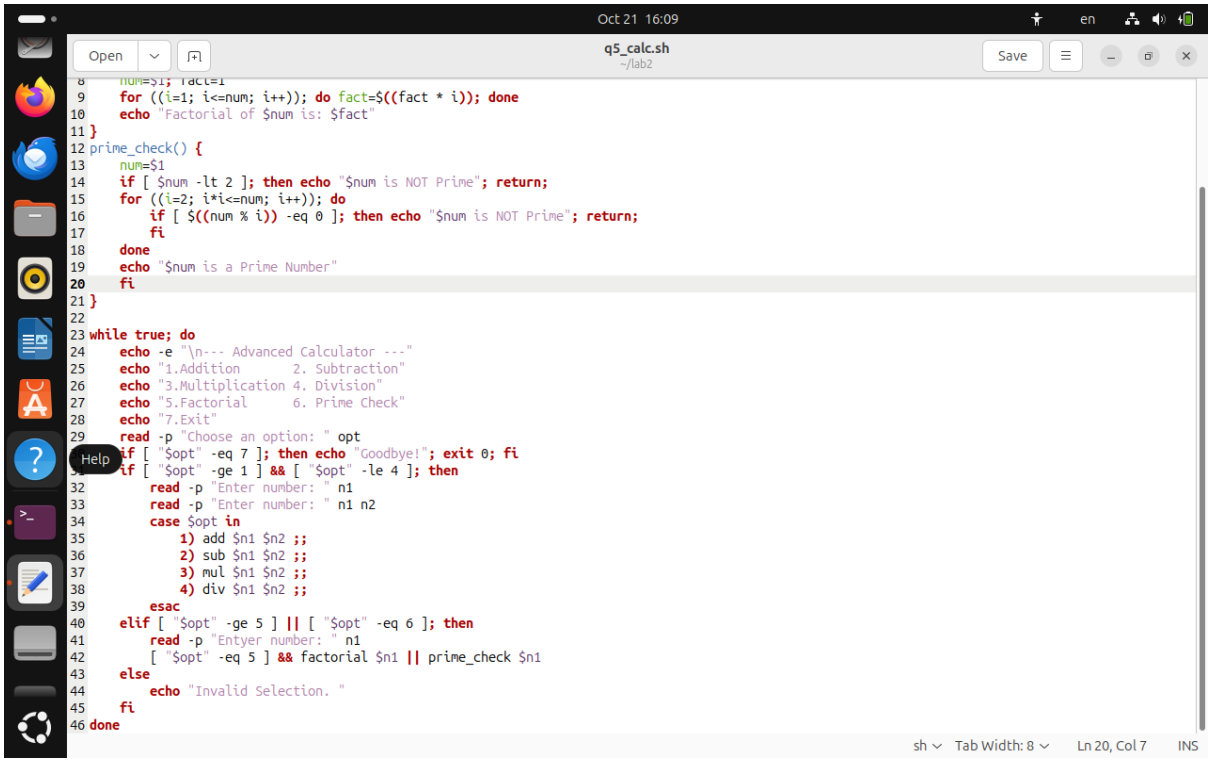
Ques5-Calculator Application Using Shell Functions.




```
#!/bin/bash
add() { echo "Result: $(($1 + $2))"; }
sub() { echo "Result: $(($1 - $2))"; }
mul() { echo "Result: $(($1 * $2))"; }
div() { echo "Result: $( echo "scale=2; $1 / $2" | bc)"; }

factorial() {
    num=$1; fact=1
    for ((i=1; i<=num; i++)); do fact=$((fact * i)); done
    echo "Factorial of $num is: $fact"
}
prime_check() {
    num=$1
    if [ $num -lt 2 ]; then echo "$num is NOT Prime"; return;
    for ((i=2; i*i<=num; i++)); do
        if [ $((num % i)) -eq 0 ]; then echo "$num is NOT Prime"; return;
        fi
    done
    echo "$num is a Prime Number"
    fi
}

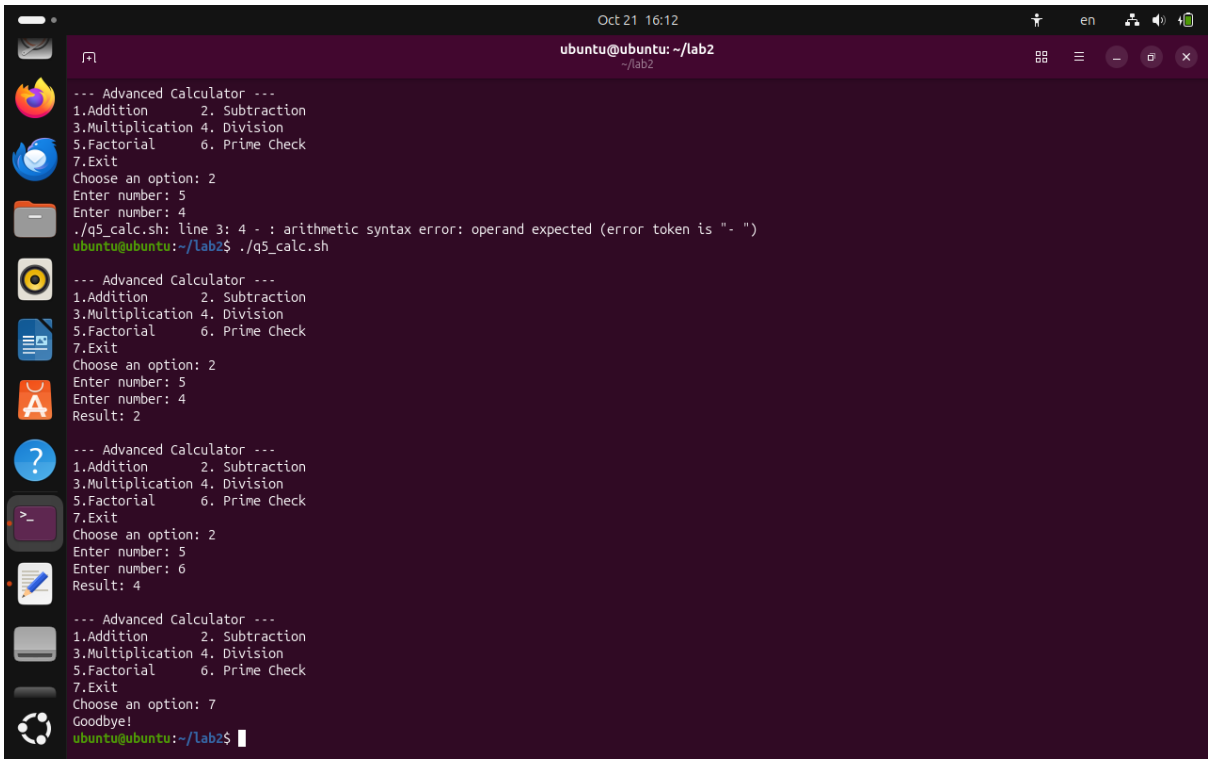
while true; do
    echo -e "\n--- Advanced Calculator ---"
    echo "1.Addition       2. Subtraction"
    echo "3.Multiplication 4. Division"
    echo "5.Factorial      6. Prime Check"
    echo "7.Exit"
    read -p "Choose an option: " opt
    if [ "$opt" -eq 7 ]; then echo "Goodbye!"; exit 0; fi
    if [ "$opt" -ge 1 ] && [ "$opt" -le 4 ]; then
        read -p "Enter number: " n1
        read -p "Enter number: " n1 n2
        case $opt in
            1) add $n1 $n2 ;;
            2) sub $n1 $n2 ;;
            3) mul $n1 $n2 ;;
            4) div $n1 $n2 ;;
        esac
```




```
    num=$1; fact=1
    for ((i=1; i<=num; i++)); do fact=$((fact * i)); done
    echo "Factorial of $num is: $fact"
}
prime_check() {
    num=$1
    if [ $num -lt 2 ]; then echo "$num is NOT Prime"; return;
    for ((i=2; i*i<=num; i++)); do
        if [ $((num % i)) -eq 0 ]; then echo "$num is NOT Prime"; return;
        fi
    done
    echo "$num is a Prime Number"
    fi
}

while true; do
    echo -e "\n--- Advanced Calculator ---"
    echo "1.Addition       2. Subtraction"
    echo "3.Multiplication 4. Division"
    echo "5.Factorial      6. Prime Check"
    echo "7.Exit"
    read -p "Choose an option: " opt
    if [ "$opt" -eq 7 ]; then echo "Goodbye!"; exit 0; fi
    if [ "$opt" -ge 1 ] && [ "$opt" -le 4 ]; then
        read -p "Enter number: " n1
        read -p "Enter number: " n1 n2
        case $opt in
            1) add $n1 $n2 ;;
            2) sub $n1 $n2 ;;
            3) mul $n1 $n2 ;;
            4) div $n1 $n2 ;;
        esac
    elif [ "$opt" -ge 5 ] || [ "$opt" -eq 6 ]; then
        read -p "Entyer number: " n1
        [ "$opt" -eq 5 ] && factorial $n1 || prime_check $n1
    else
        echo "Invalid Selection. "
    fi
done
```

Output:-


Oct 21 16:12
en
ubuntu@ubuntu: ~/lab2
~/lab2
--- Advanced Calculator ---
1.Addition 2. Subtraction
3.Multiplication 4. Division
5.Factorial 6. Prime Check
7.Exit
Choose an option: 2
Enter number: 5
Enter number: 4
./q5_calc.sh: line 3: 4 - : arithmetic syntax error: operand expected (error token is "- ")
ubuntu@ubuntu:~/lab2$ ./q5_calc.sh
--- Advanced Calculator ---
1.Addition 2. Subtraction
3.Multiplication 4. Division
5.Factorial 6. Prime Check
7.Exit
Choose an option: 2
Enter number: 5
Enter number: 4
Result: 2
--- Advanced Calculator ---
1.Addition 2. Subtraction
3.Multiplication 4. Division
5.Factorial 6. Prime Check
7.Exit
Choose an option: 2
Enter number: 5
Enter number: 6
Result: 4
--- Advanced Calculator ---
1.Addition 2. Subtraction
3.Multiplication 4. Division
5.Factorial 6. Prime Check
7.Exit
Choose an option: 7
Goodbye!
ubuntu@ubuntu:~/lab2$